

# MediCare Clinic Appointment System Documentation

This document provides a comprehensive overview of the MediCare Clinic Appointment System, detailing the idea and its structure, features, technologies, and system flow.

**Deployed Browser-Based Application:** <https://medicare-booking-sage.vercel.app/>

## Team Members:

| Name                       | ID       |
|----------------------------|----------|
| Omar Osama Mohamed         | 22101414 |
| Nour Mohamed El Barawy     | 22100165 |
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| Mariam Mahmoud Abdelmoneim | 22100808 |

## 1. Project Idea

### 1.1 Project Vision

The **MediCare Clinic Appointment System** was conceived as a comprehensive digital solution to bridge the gap between healthcare providers and patients. The core idea was to replace traditional, manual scheduling with a robust, **Client-Server web application** that handles the complete lifecycle of a medical appointment—from patient registration and specialist discovery to real-time booking and doctor-led schedule management.

### 1.2 Core Features

The system is designed to provide a tailored experience for two distinct user roles:

- **For Patients:** A seamless flow to browse specialists, check real-time availability based on the doctor's specific working days, and manage their own medical schedule through a private dashboard.
- **For Doctors:** A professional portal that summarizes upcoming patient loads and allows for the administrative cancellation of appointments, ensuring the clinic's schedule remains accurate.

## 2. Technologies Used & System Architecture

### 2.1 Technical Stack Overview

The system is a sophisticated **React Single Page Application (SPA)** designed to streamline healthcare scheduling. By moving away from traditional multi-page navigation, this project provides a fluid, app-like experience where users can register, book appointments, and manage schedules without disruptive page reloads.

| Technology           | Role in Project  | Key Implementation Detail                                                                                                               |
|----------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| React.js             | Core Framework   | Defines the semantic layout of the five primary web pages: Home, Login, Booking, and the two Dashboards.                                |
| React Router         | Routing          | Manages the browser history and URL paths (/login, /booking, etc.) without page refreshes.                                              |
| Context API          | State Management | Centralizes authentication logic and user metadata via AuthContext.                                                                     |
| Node.js & Express    | Backend Server   | Acts as the middleware that handles RESTful API routing and handles authentication and communication between the user and the database. |
| SQLite               | Database         | Persistent relational storage for patients, doctors, and appointment records.                                                           |
| JWT (JSON Web Token) | Authentication   | Enables secure, stateless user sessions by storing an encrypted token in the browser's localStorage.                                    |
| Bcrypt               | Security         | Used to hash and salt user passwords before they are saved to the database, ensuring sensitive data is protected.                       |

|                                         |                           |                                                                                                                                       |
|-----------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Web App Manifest</b>                 | PWA Configuration         | A JSON file that defines the app's name, theme colors, and the <code>display: standalone</code> property to remove browser UI.        |
| <b>PWA (Progressive Web App)</b>        | Native Experience         | Allows the website to be installed on devices, featuring a custom icon and standalone window launching.                               |
| <b>Vercel</b>                           | Frontend Hosting          | Deploys the React SPA to a global CDN for high availability and fast performance.                                                     |
| <b>Render</b>                           | Backend Hosting           | Hosts the live Node.js API and SQLite database, allowing the application to be accessed via a public URL rather than localhost.       |
| <b>CORS &amp; Environment Variables</b> | Cross-Origin Connectivity | Configured to allow secure communication between the Vercel frontend and the Render backend link, replacing port 5050 for production. |

## 2.2 Backend & Database Architecture

The transition to a real backend was a critical architectural decision to move beyond temporary storage and achieve a functional, multi-user system.

### A. Backend Setup

- Node.js was used to build the backend server
- Express framework was used for routing
- CORS enabled to allow frontend communication
- Environment variables managed using `.env`

### B. Relational Database Design

The system relies on an SQLite database with three interconnected tables to maintain data integrity:

- **Doctors Table:** Stores professional details, encrypted credentials, and availability JSON strings.

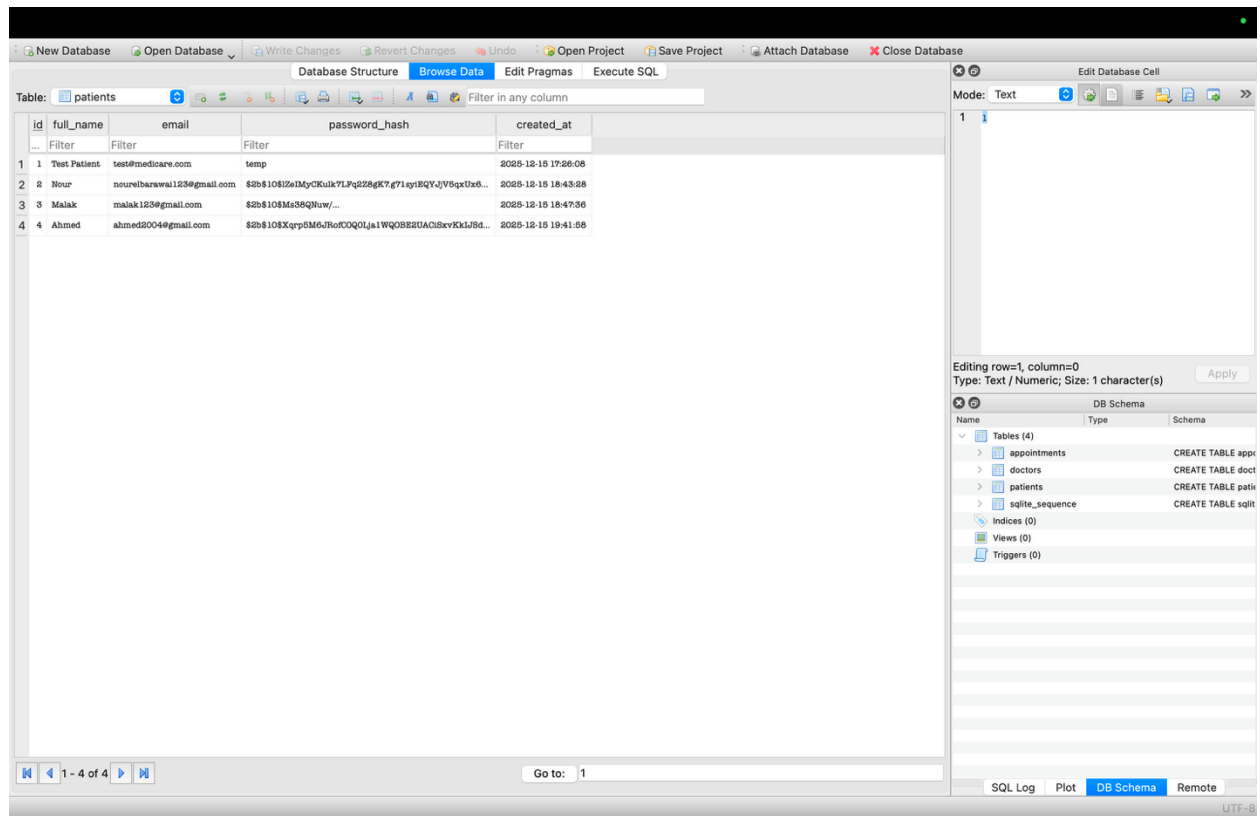
| Column        | Description        |
|---------------|--------------------|
| id            | Unique doctor ID   |
| full_name     | Doctor name        |
| specialty     | Medical specialty  |
| days_json     | Available days     |
| days_text     | Readable schedule  |
| email         | Doctor email       |
| password_hash | Encrypted password |

The screenshot shows a database management interface with the 'doctors' table selected. The table has 8 columns: id, full\_name, specialty, days\_json, days\_text, email, password\_hash, and created\_at. The data is displayed in a grid with 4 rows. The right sidebar shows the database schema, including tables (appointments, doctors, patients, sqlite\_sequence), indices, views, and triggers.

| id | full_name          | specialty      | days_json   | days_text     | email                 | password_hash                                      | created_at          |
|----|--------------------|----------------|-------------|---------------|-----------------------|----------------------------------------------------|---------------------|
| 1  | Dr. Omar Osama     | Cardiologist   | [1,3,5]     | Mon, Wed, Fri | dromar@medicare.com   | \$2b\$10\$u0.3u1VM968h7M7oLaT2t.gH6nISXtW4Y36gJ... | 2025-12-15 16:59:04 |
| 2  | Dr. Yousef Mohamed | Dermatologist  | [2,4]       | Tue, Thu      | dryousef@medicare.com | \$2b\$10\$pmDjGa7LYh3jGDvGdMYuG8cVoyP8epAq5R.w...  | 2025-12-15 16:59:04 |
| 3  | Dr. Nour Mohamed   | Pediatrician   | [0,6]       | Sun, Sat      | drnour@medicare.com   | \$2b\$10\$1PB8C80TyTh22W20vtqj4AkYeVaJBNTVoPRK...  | 2025-12-15 16:59:04 |
| 4  | Dr. Mariam Mahmoud | General Doctor | [0,1,2,3,4] | Sun to Thu    | drmariam@medicare.com | \$2b\$10\$4inP8CF88pV8lQ7a5Bq8x0J73bU0EvudP08KL... | 2025-12-15 16:59:04 |

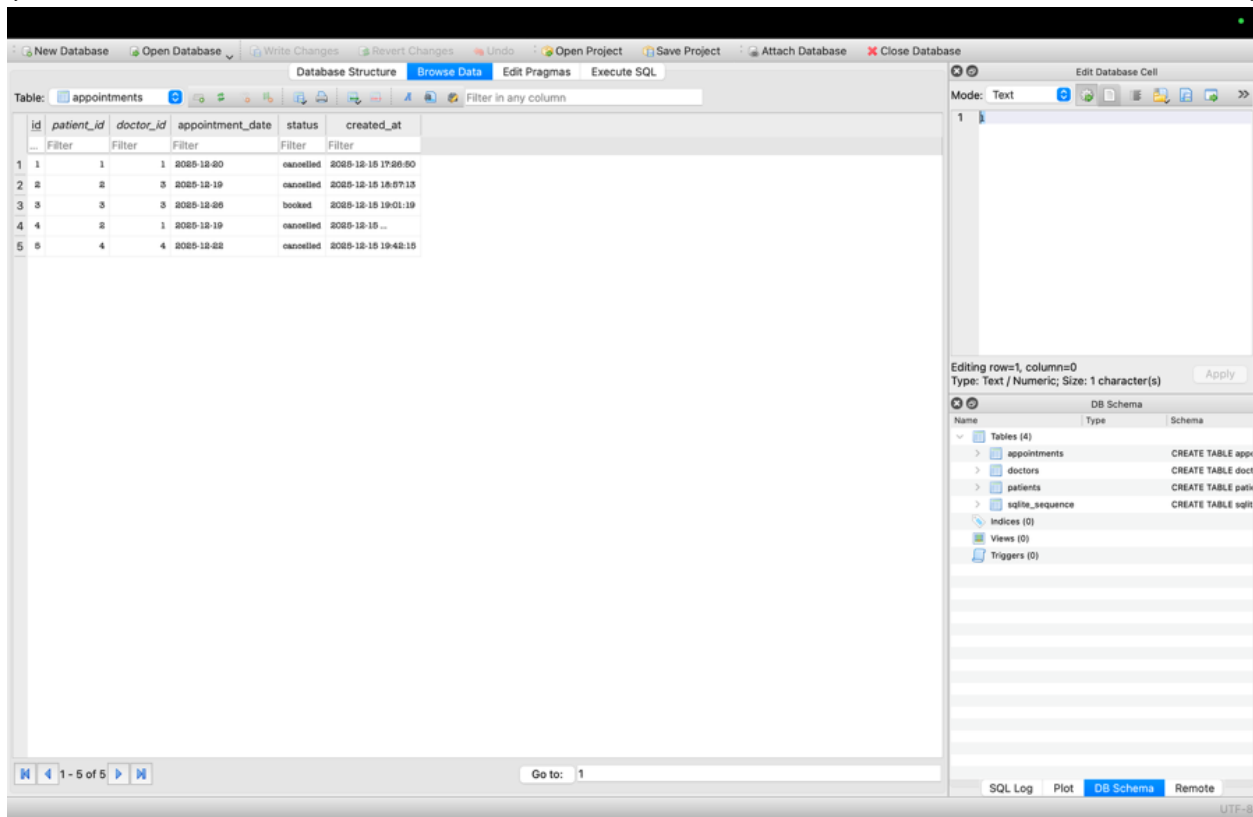
- **Patients Table:** Stores registered user profiles and encrypted passwords.

| Column        | Description        |
|---------------|--------------------|
| id            | Patient ID         |
| full_name     | Patient name       |
| email         | Patient email      |
| password_hash | Encrypted password |



- **Appointments Table:** Acts as a junction table linking `patient_id` and `doctor_id` with specific dates and statuses.

| Column           | Description        |
|------------------|--------------------|
| id               | Appointment ID     |
| doctor_id        | Linked doctor      |
| patient_id       | Linked patient     |
| appointment_date | Appointment date   |
| status           | booked / cancelled |



### C. RESTful API & Communication

Communication between the frontend and backend is handled via standard HTTP methods (GET, POST, DELETE).

- **Data Exchange:** All data is transmitted in **JSON** format to ensure compatibility between the JavaScript frontend and the Node.js backend.
- **Role-Based Access Control (RBAC):** Backend middleware (`auth.js`) validates the JWT for every request to ensure that patients can only see their own appointments and doctors can only access their specific portals.

## ***D. Global Session Management***

A major architectural choice was the implementation of the AuthProvider component. This component:

- **Persists Login:** Checks `localStorage` on initial load to restore the user's session (token and `currentUser`).
- **Centralizes Logout:** Provides a single `logout()` function that clears storage and updates the global state simultaneously to protect private routes.

## ***3. Implementation Steps & System Flow***

### **3.1 Development Phases**

The project was completed in three main phases:

#### **Phase 1: Front-End Design**

Designed a user-friendly and responsive interface using HTML, CSS, and JavaScript, focusing on usability and accessibility across devices.

#### **Phase 2: Backend Integration & React**

Converted the front-end into a React SPA using React Router and Context API, and implemented a Node.js/Express backend connected to an SQLite database for managing users and appointments.

#### **Phase 3: Finalization, Browser-Based Conversion, and Documentation**

Converted the website into a fully browser-based application with app-like behavior, conducted final testing of all CRUD operations to ensure system stability, prepared comprehensive project documentation, and collaborated as a team to present and demonstrate the final application.

#### **Browser-Based Application:**

To fulfill the bonus requirements for a technically advanced approach, the system was transformed into a **Progressive Web App (PWA)**.

### ***1. Single Page Application (SPA) Architecture***

Using **React Router**, the app functions as a single page. Navigation between the Home, Booking, and Dashboards is instantaneous, eliminating page reloads and providing a desktop-app feel.

### ***2. Standalone Display & Manifest Integration***

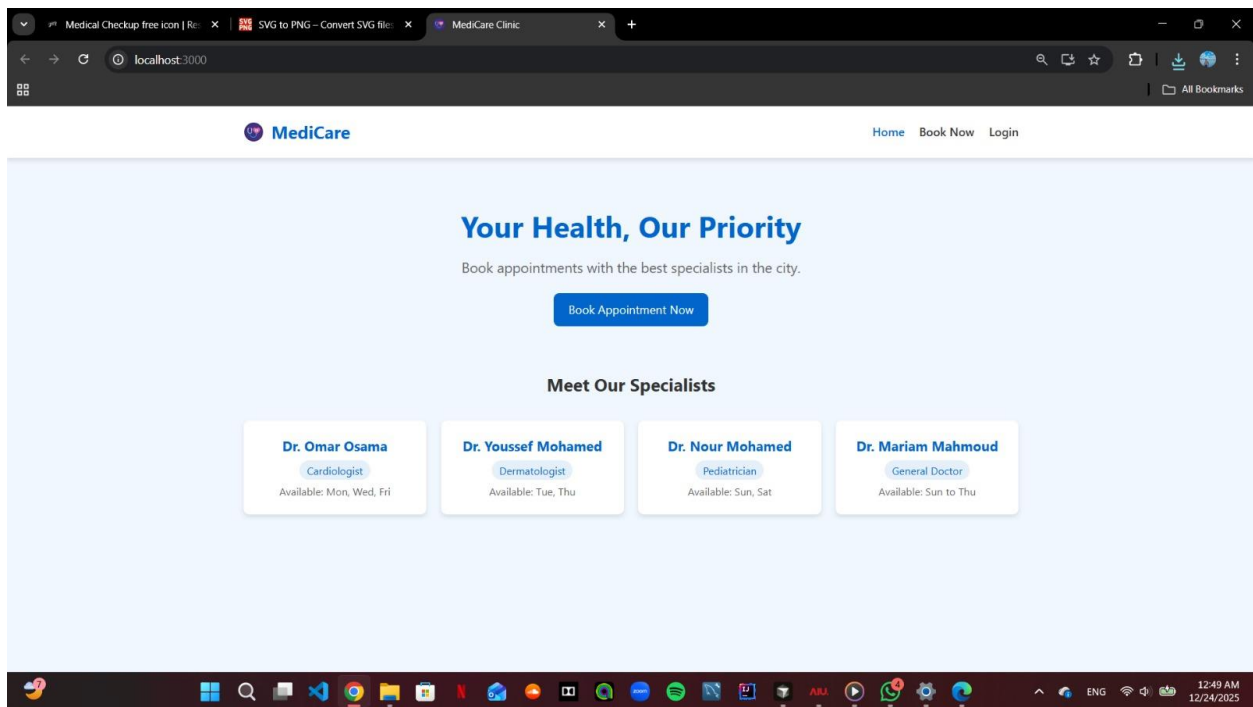
A **Web App Manifest** was implemented to define app behavior upon installation:

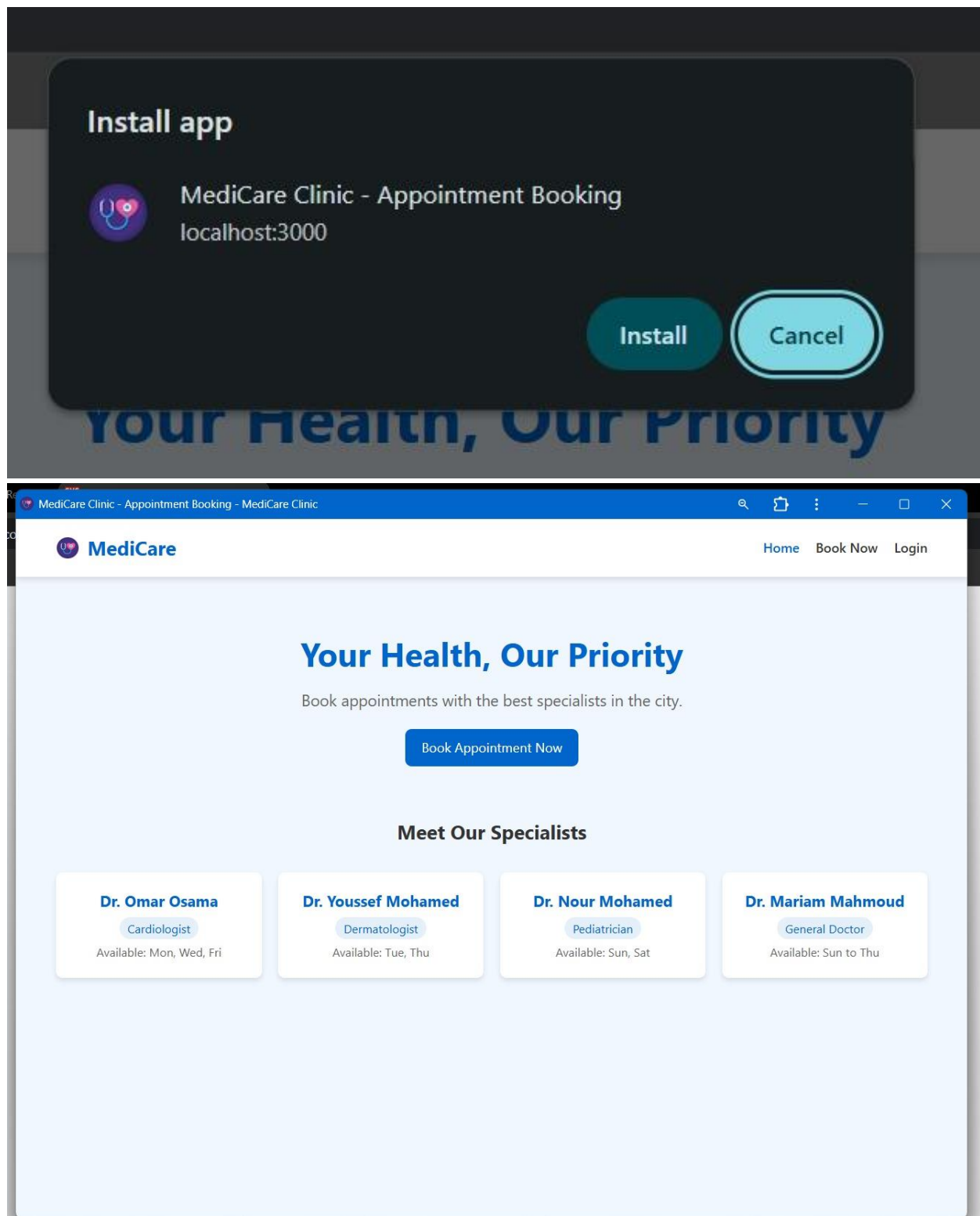
- **Standalone Mode:** By setting `display: standalone`, the app launches in its own window without browser UI elements (address bars), mimicking a native application.
- **Custom Branding:** Includes a unique **MediCare Icon** for "Add to Home Screen" functionality.

### ***3. Cloud Deployment & Connectivity***

The project was moved from a local environment to a live production state:

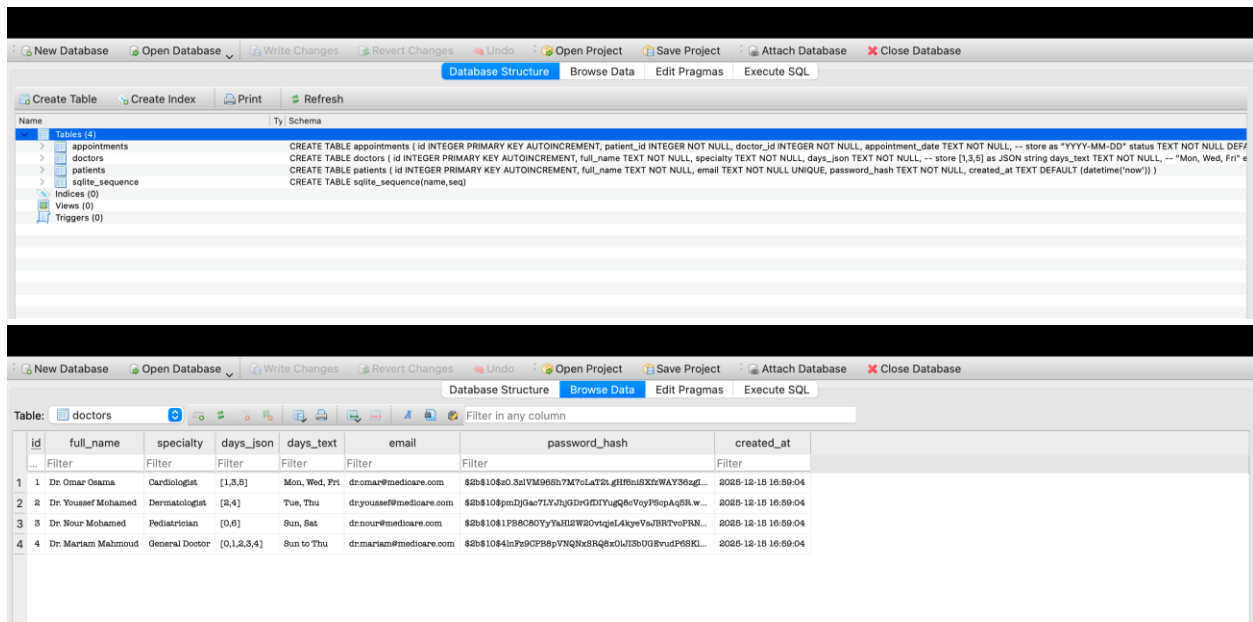
- **Hosting:** The React frontend is hosted on **Vercel**, while the Node.js API and SQLite database are deployed on **Render**.
- **Connectivity:** Secured via **HTTPS** and **CORS** configurations, allowing the Vercel PWA to safely communicate with the Render backend.



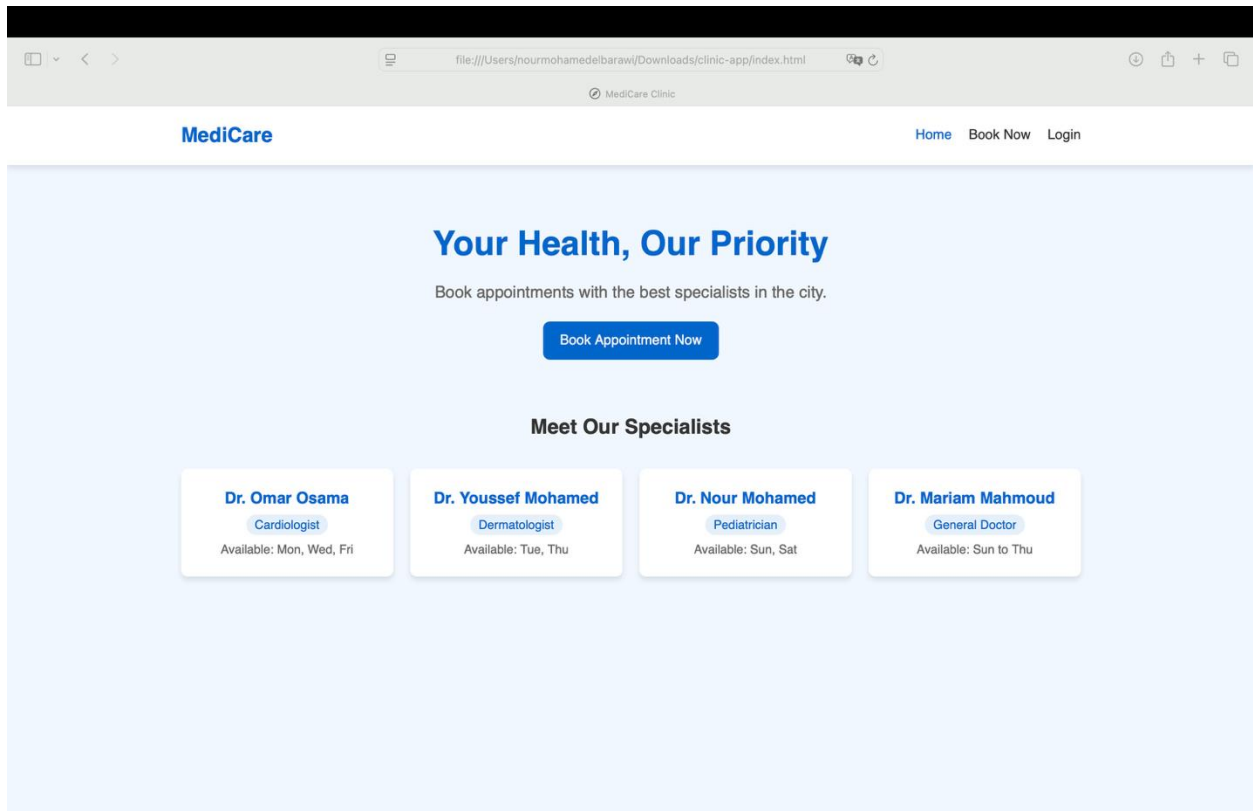


### 3.2 System Flow

#### Database Initialization & Seeding:



## Home Page:



## A. Authentication Flow

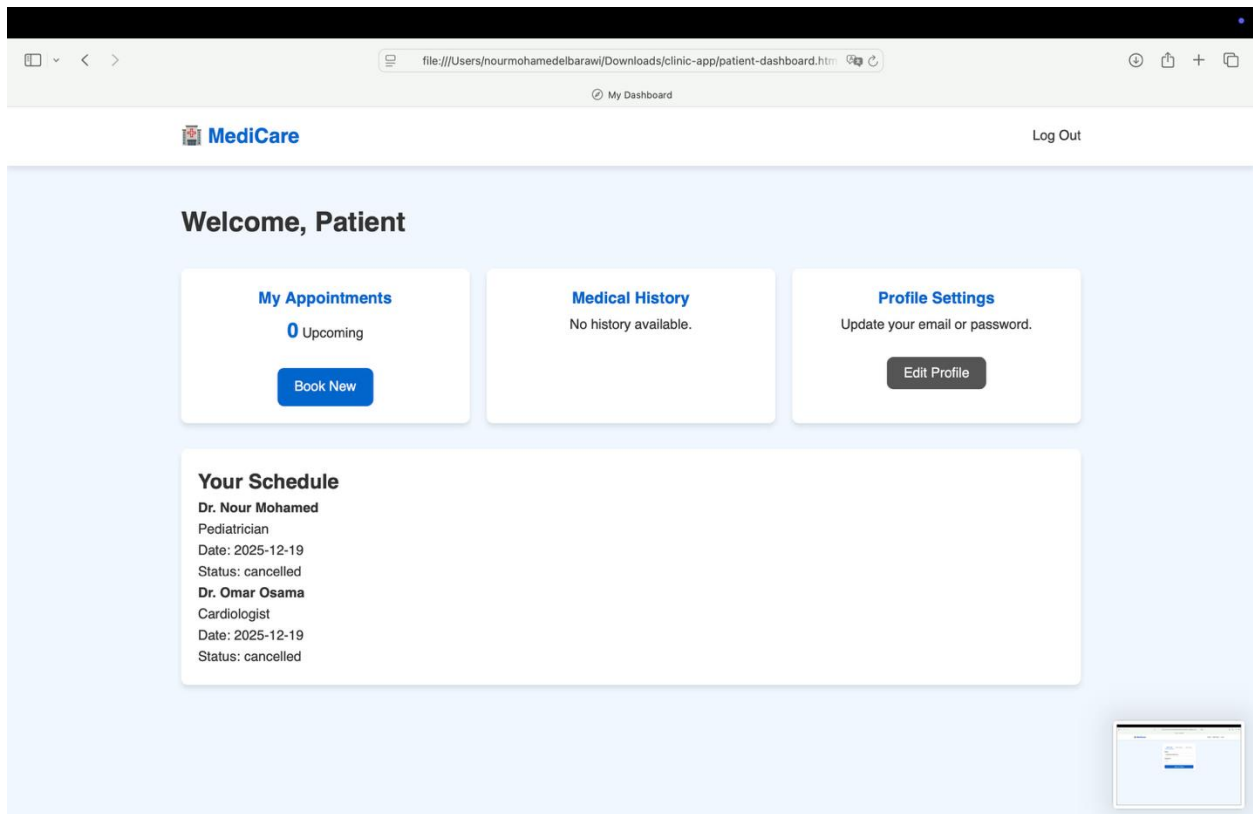
1. The user enters credentials into the LoginPage component.

2. React sends a POST request to the Node.js backend (/api/auth/patient/login or /api/auth/doctor/login).
3. The backend verifies the user against the SQLite database using **Bcrypt** decryption.
4. Upon success, the server returns a **JWT token** and user metadata.
5. The AuthProvider executes `saveSession`, updating the global state and storing the token in `localStorage`.

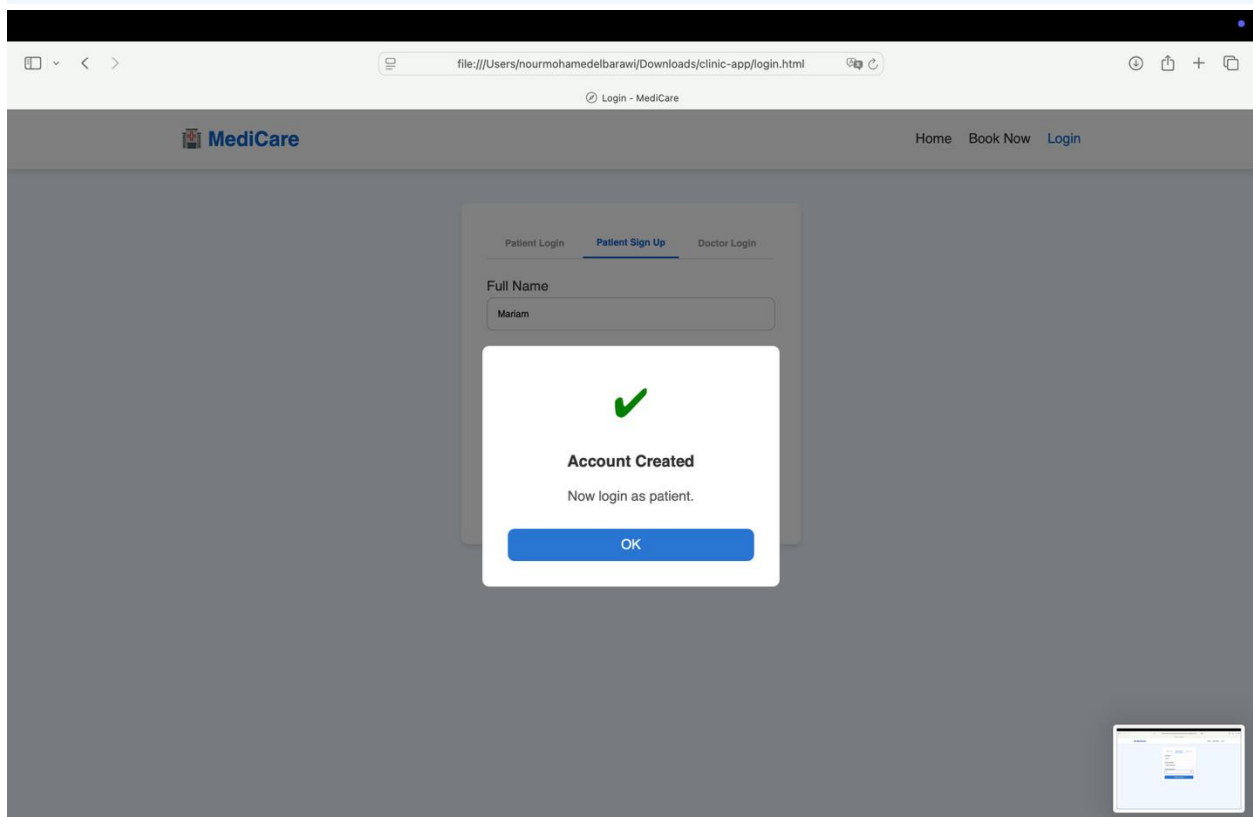
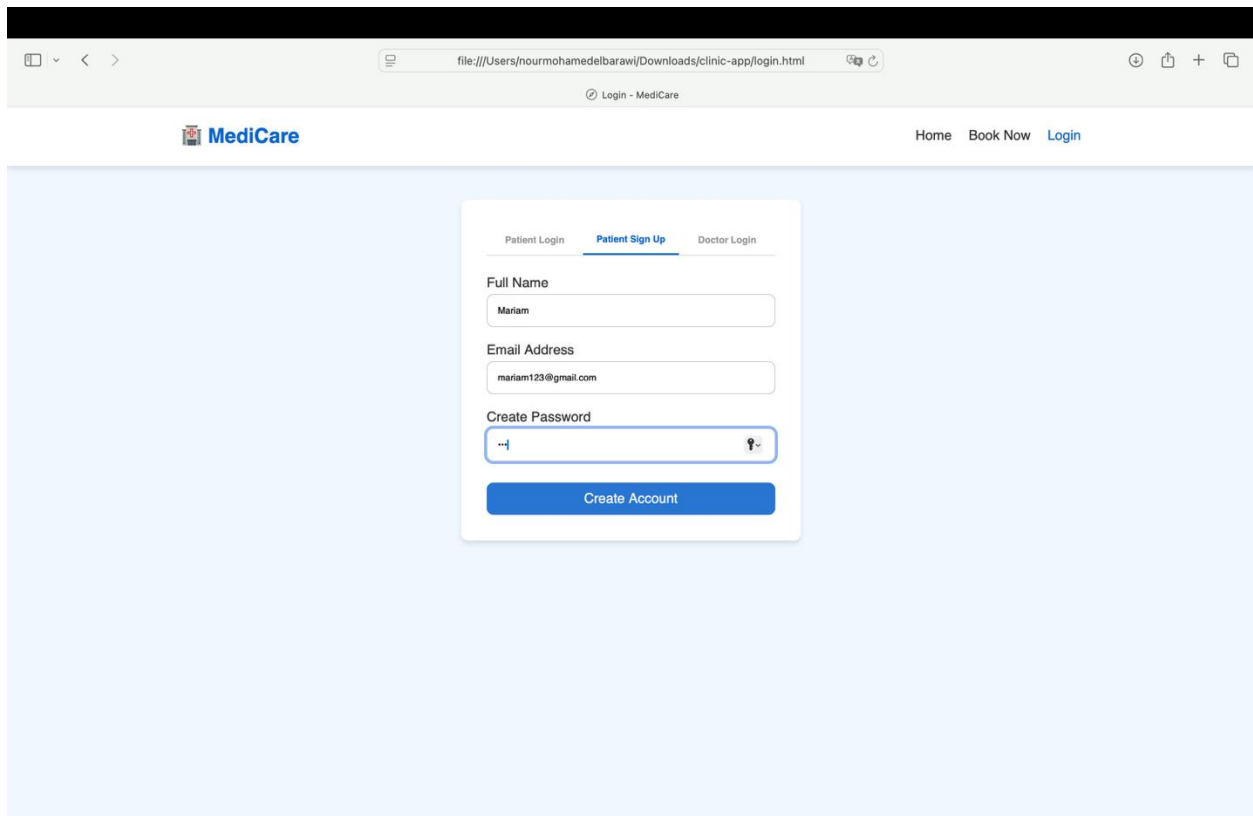
## Patient Login:

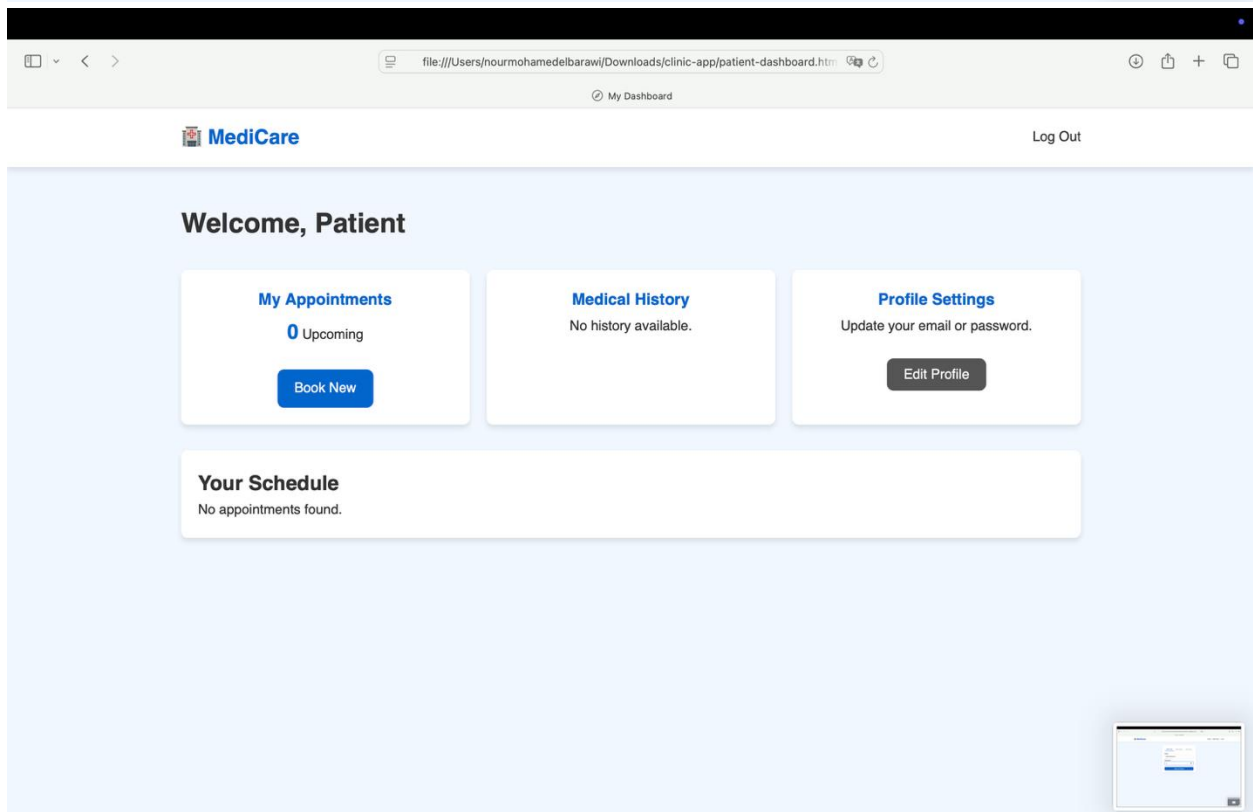
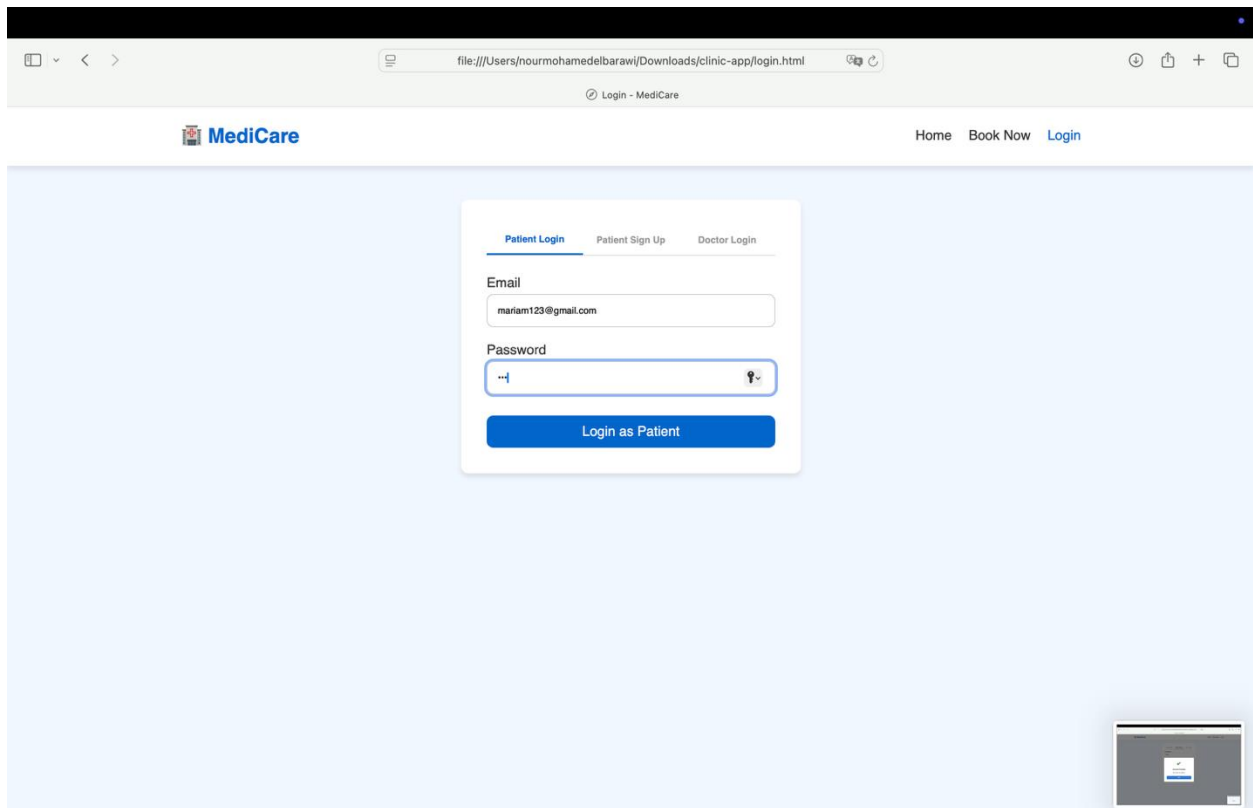
The screenshot shows a web browser window displaying the 'MediCare' application. The browser's address bar shows the file path: `file:///Users/hourmohamedelbarawi/Downloads/clinic-app/login.html`. The page has a header with the 'MediCare' logo and navigation links: 'Home', 'Book Now', and 'Login'. The main content area features a light blue background with a central white login form. The form has three tabs: 'Patient Login' (selected), 'Patient Sign Up', and 'Doctor Login'. It contains two input fields: 'Email' with the value 'nourelbarawai123@gmail.com' and 'Password' with masked characters '\*\*\*\*\*'. A blue button labeled 'Login as Patient' is positioned below the password field.

## Patient's Dashboard:



**Patient Registration (Sign-Up):**





| New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database |              |                     |                                                    |                     |  |
|-----------------------------------------------------------------------------------------------------------------------|--------------|---------------------|----------------------------------------------------|---------------------|--|
| Database Structure Browse Data Edit Pragma Execute SQL                                                                |              |                     |                                                    |                     |  |
| Table: patients Filter in any column                                                                                  |              |                     |                                                    |                     |  |
| id                                                                                                                    | full_name    | email               | password_hash                                      | created_at          |  |
| ...                                                                                                                   | Filter       | Filter              | Filter                                             | Filter              |  |
| 1                                                                                                                     | Test Patient | test@medicare.com   | temp                                               | 2025-12-15 17:26:08 |  |
| 2                                                                                                                     | Nour         | nour@medicare.com   | \$2b\$10\$ZxIMyCKuik7LPq22gK7g71eytBQVQjVbqx0x8... | 2025-12-15 18:43:28 |  |
| 3                                                                                                                     | Malak        | malak123@gmail.com  | \$2b\$10\$Mz09QWuw/...                             | 2025-12-15 18:47:36 |  |
| 4                                                                                                                     | Ahmed        | ahmed2004@gmail.com | \$2b\$10\$Xqrp5M67hoCOQ0Lja1WQ0BIEUACisxvYKd18d... | 2025-12-15 19:41:56 |  |
| 5                                                                                                                     | Mariam       | maria123@gmail.com  | \$2b\$10\$KJ2x94sd1h90VfB2wst7gR8d5JrQMxv00K...    | 2025-12-15 20:20:18 |  |

Doctor Login:

file:///Users/hourmohamedelbarawi/Downloads/clinic-app/login.html

Login - MediCare

MediCare

HomeBook NowLogin

Patient LoginPatient Sign UpDoctor Login

Doctor Email

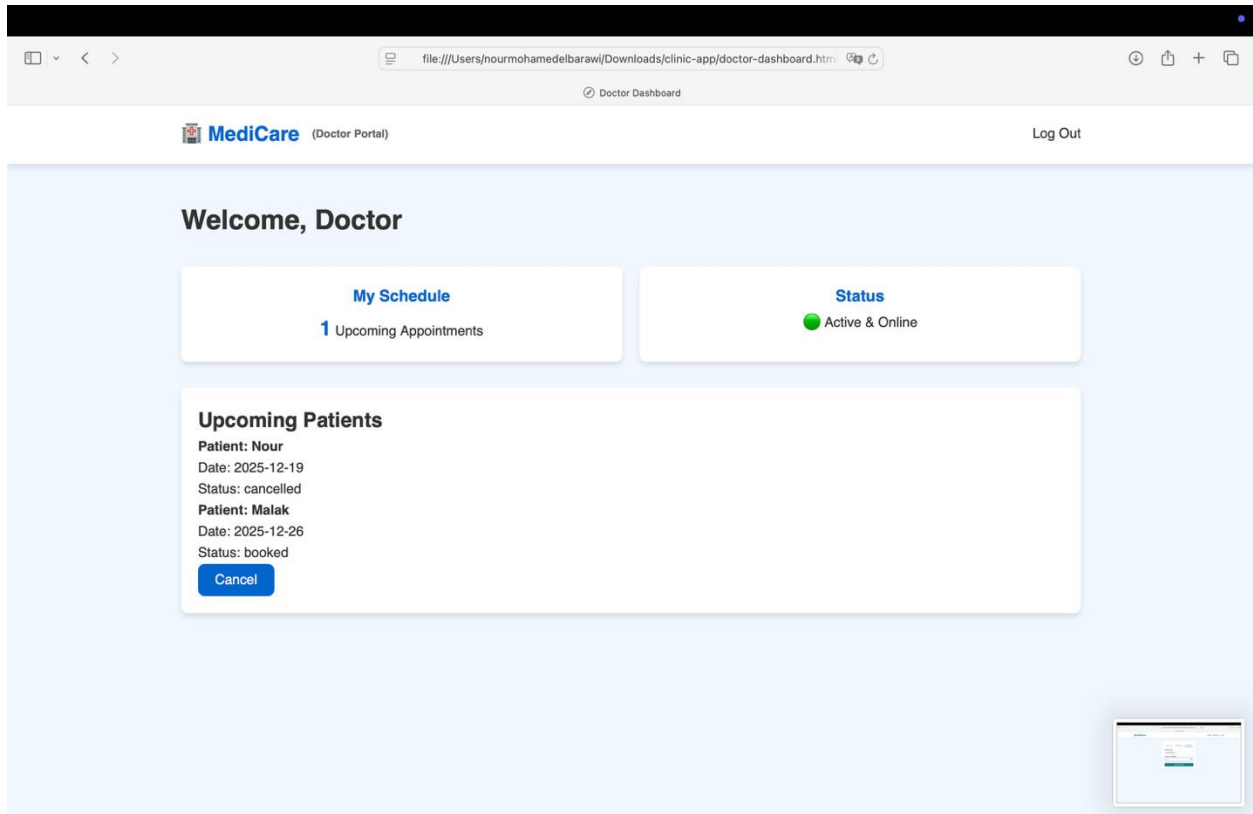
dr.nour@medicare.com

License / Password

.....

Login as Doctor

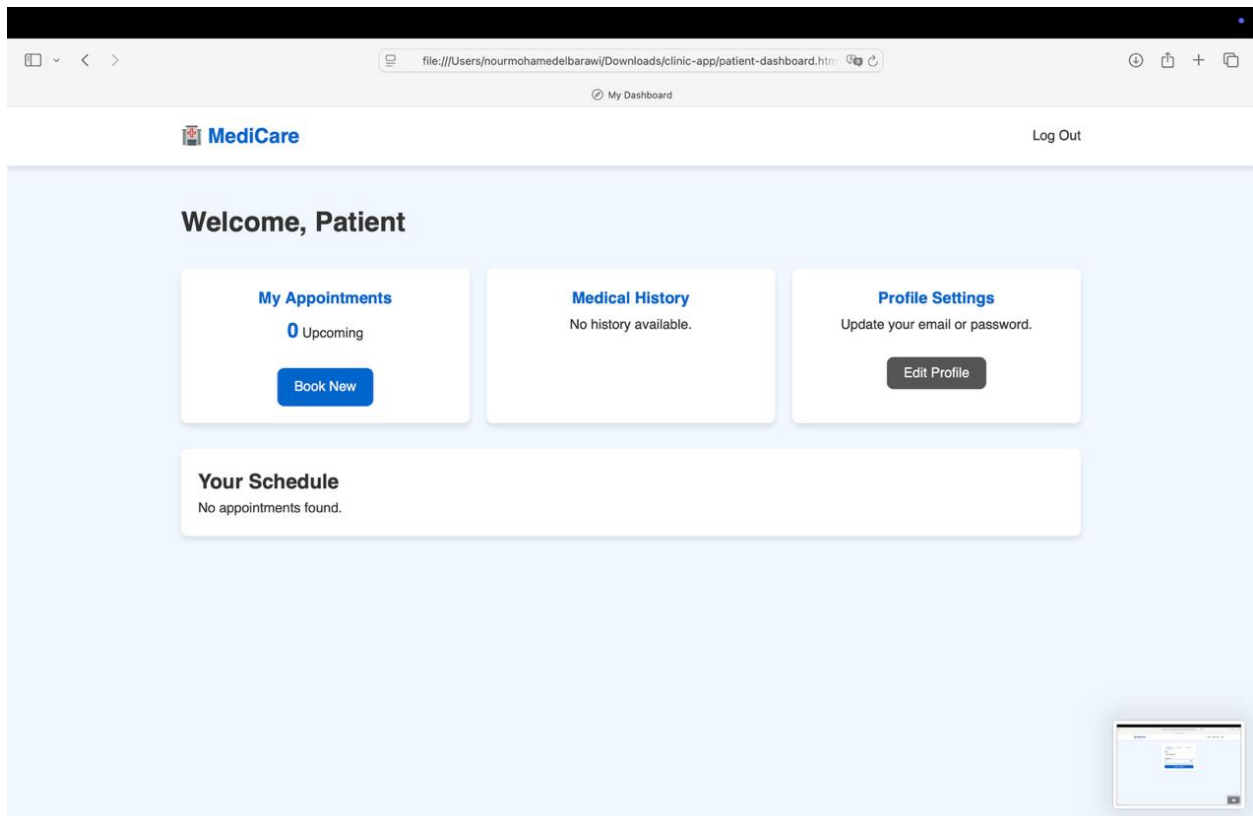
Doctor’s Dashboard:



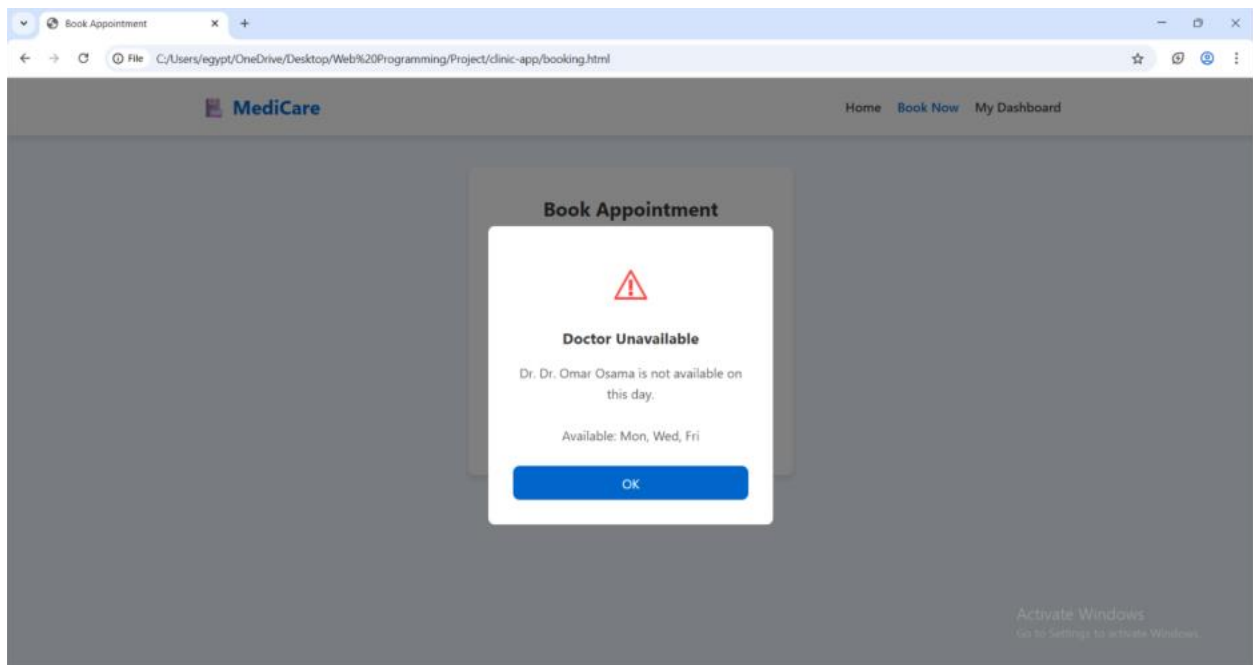
## B. Booking Logic Flow

1. The BookingPage fetches the current doctor list and availability from the database.
2. When a user selects a date, the React state performs an **immediate client-side validation**.
3. The system checks the selected date against the doctor's days array to verify availability (dynamic availability validation).
4. If valid, a POST request with the **Authorization Header** is sent to the backend.
5. The backend records the appointment, and the UI redirects the user to the PatientDashboard to view their updated schedule.

### Before Booking an Appointment:



## Appointment Availability Checking and Error Handling:



## After Booking an Appointment:

file:///Users/nourmohamedelbarawi/Downloads/clinic-app/booking.html

Book Appointment

MediCare

HomeBook NowMy Dashboard

Book Appointment

Logged in as: Mariam

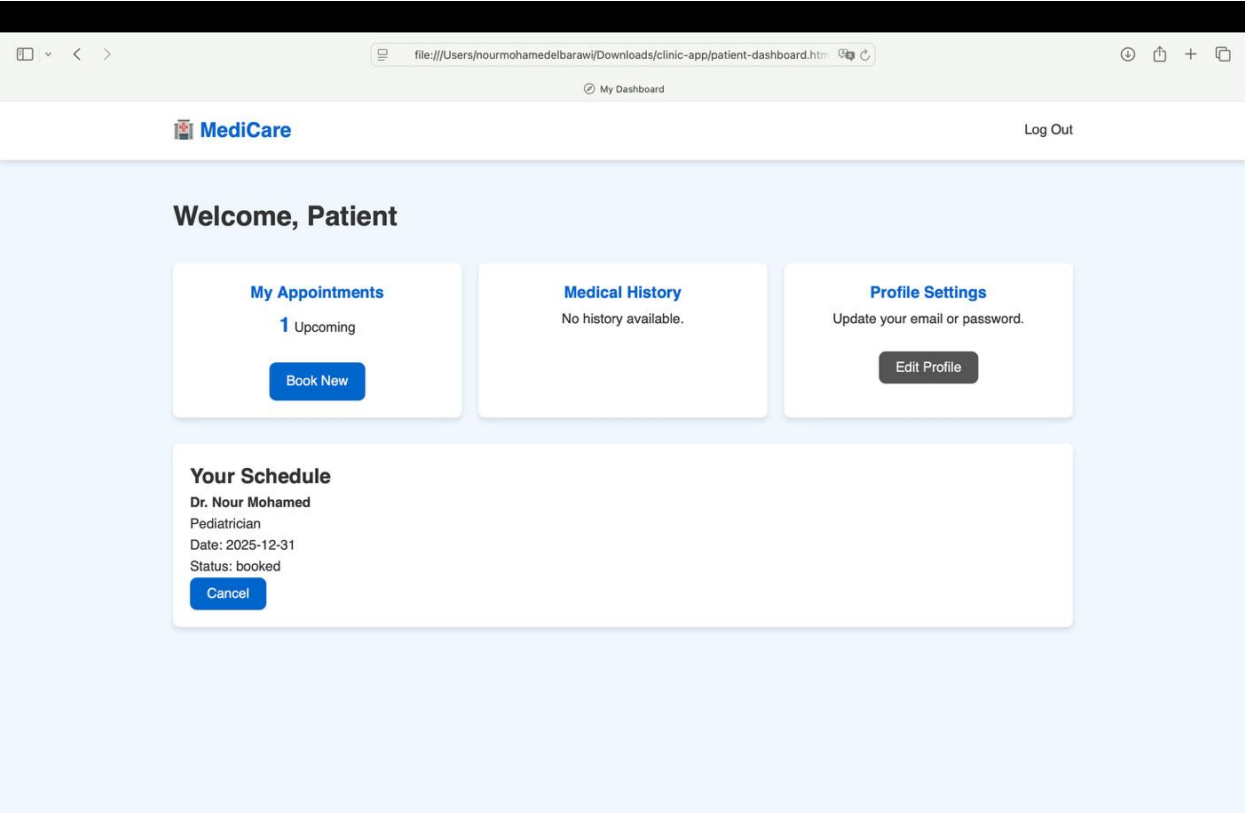
Select Doctor & Schedule

Dr. Nour Mohamed -- Pediatrician (Sun, Sat)

Select Date

31/12/2025

Confirm Booking



|                                                                                                                       |     |            |           |                  |           |                     |  |  |  |
|-----------------------------------------------------------------------------------------------------------------------|-----|------------|-----------|------------------|-----------|---------------------|--|--|--|
| New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database |     |            |           |                  |           |                     |  |  |  |
| Database Structure Browse Data Edit Pragmas Execute SQL                                                               |     |            |           |                  |           |                     |  |  |  |
| Table: appointments Filter in any column                                                                              |     |            |           |                  |           |                     |  |  |  |
|                                                                                                                       | id  | patient_id | doctor_id | appointment_date | status    | created_at          |  |  |  |
|                                                                                                                       | ... | Filter     | Filter    | Filter           | Filter    | Filter              |  |  |  |
| 1                                                                                                                     | 1   | 1          | 1         | 2025-12-20       | cancelled | 2025-12-15 17:26:50 |  |  |  |
| 2                                                                                                                     | 2   | 2          | 3         | 2025-12-19       | cancelled | 2025-12-15 18:57:13 |  |  |  |
| 3                                                                                                                     | 3   | 3          | 3         | 2025-12-26       | booked    | 2025-12-15 19:01:19 |  |  |  |
| 4                                                                                                                     | 4   | 2          | 1         | 2025-12-19       | cancelled | 2025-12-15 19:39:44 |  |  |  |
| 5                                                                                                                     | 5   | 4          | 4         | 2025-12-22       | cancelled | 2025-12-15 19:42:15 |  |  |  |
| 6                                                                                                                     | 6   | 5          | 3         | 2025-12-31       | booked    | 2025-12-15 20:23:00 |  |  |  |

## Appointment Appearing in Doctor's Dashboard:

file:///Users/nourmohamedelbarawi/Downloads/clinic-app/doctor-dashboard.html

Doctor Dashboard

**MediCare** (Doctor Portal) [Log Out](#)

### Welcome, Doctor

**My Schedule**  
2 Upcoming Appointments

**Status**  
● Active & Online

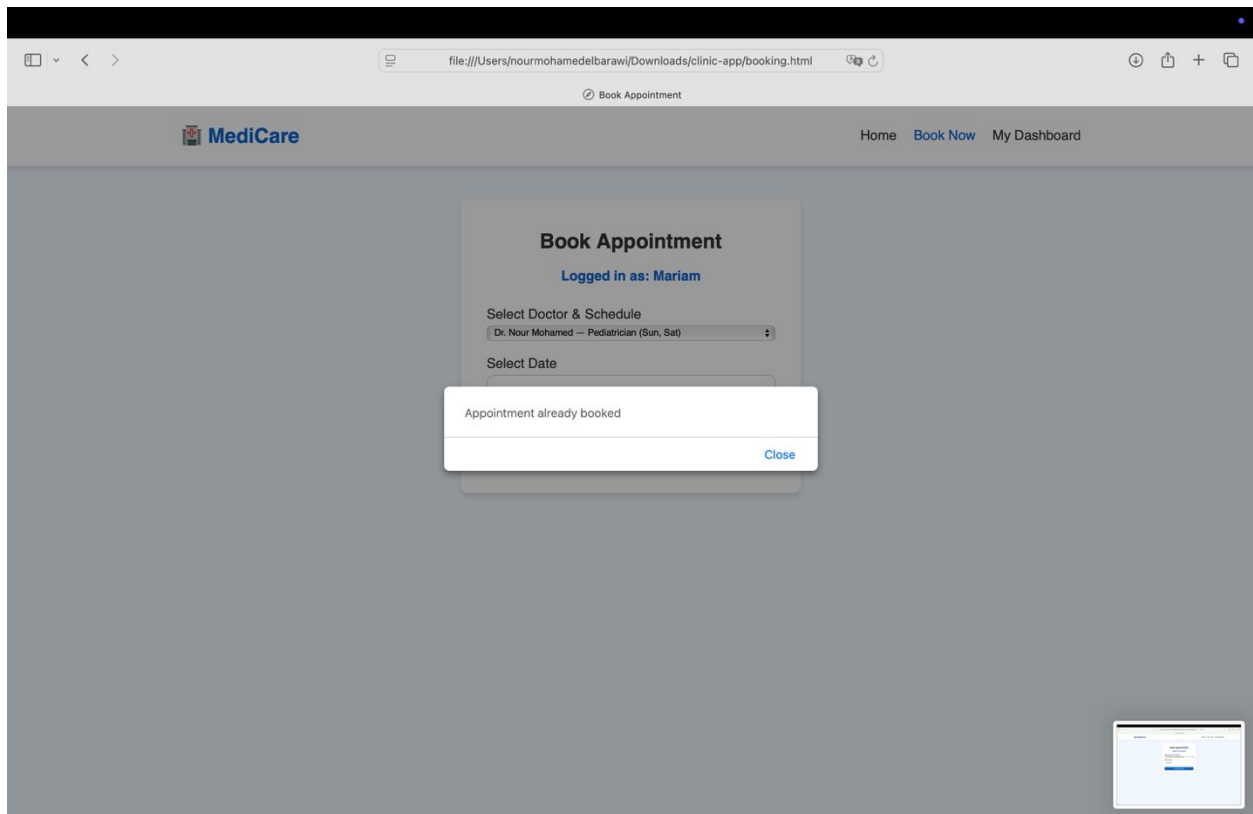
**Upcoming Patients**

**Patient: Nour**  
Date: 2025-12-19  
Status: cancelled

**Patient: Malak**  
Date: 2025-12-26  
Status: booked  
[Cancel](#)

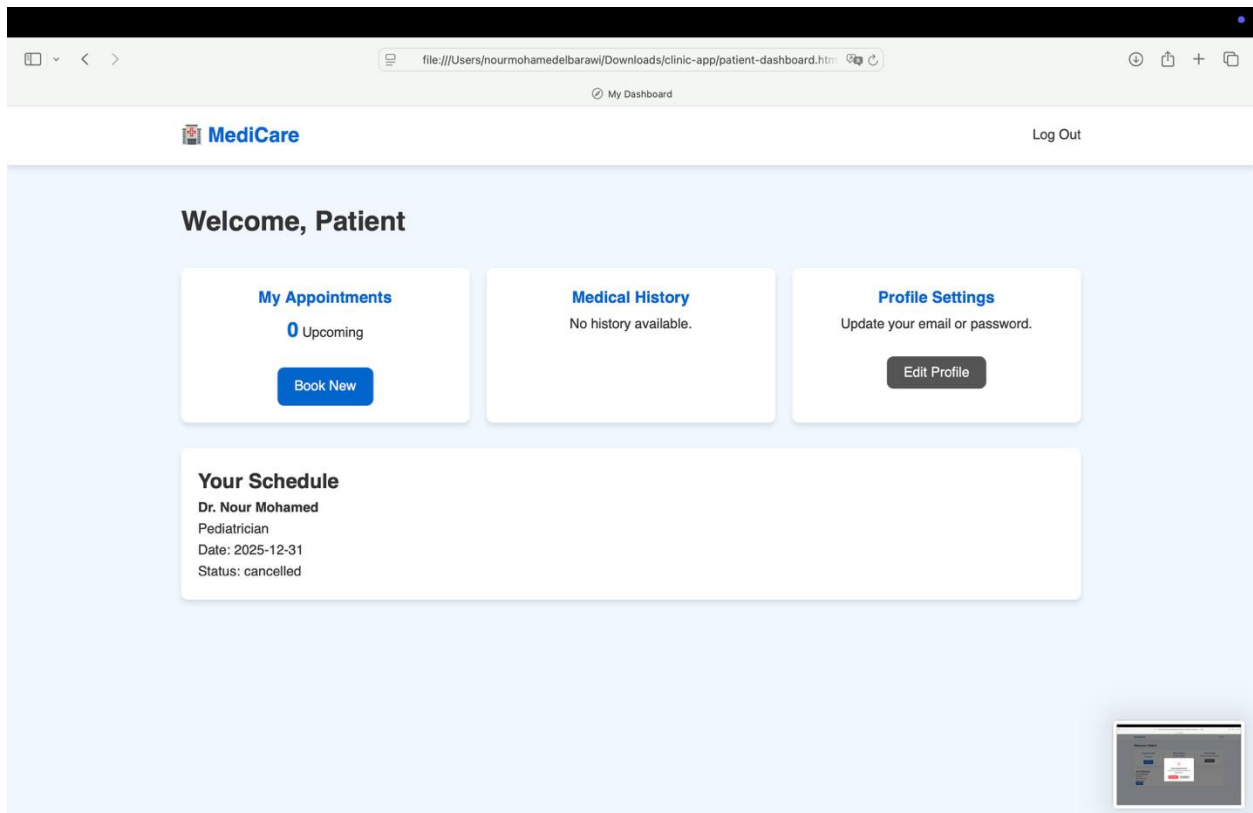
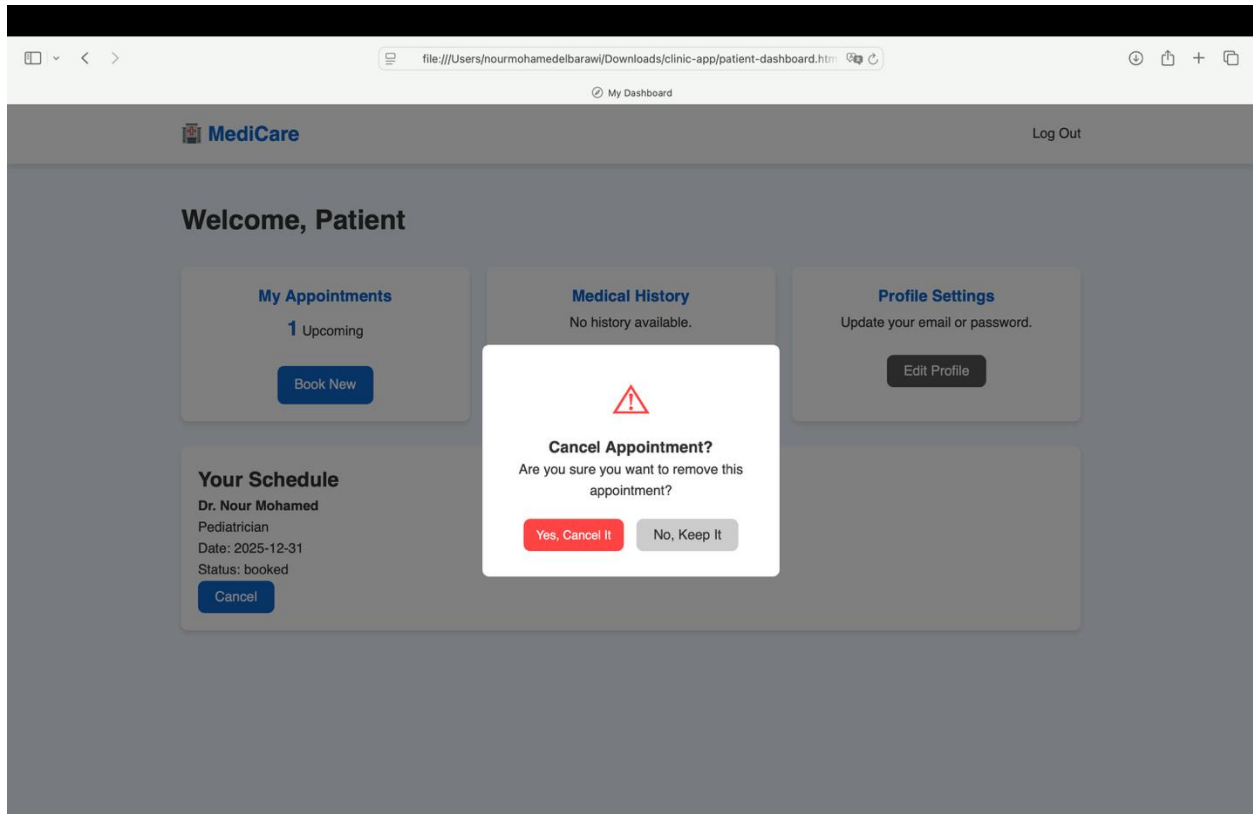
**Patient: Mariam**  
Date: 2025-12-31  
Status: booked  
[Cancel](#)

**Validation:**

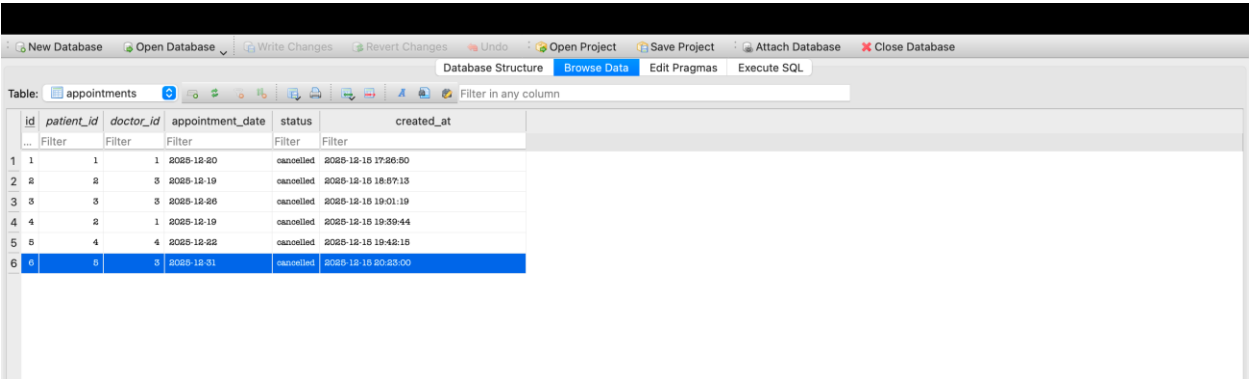
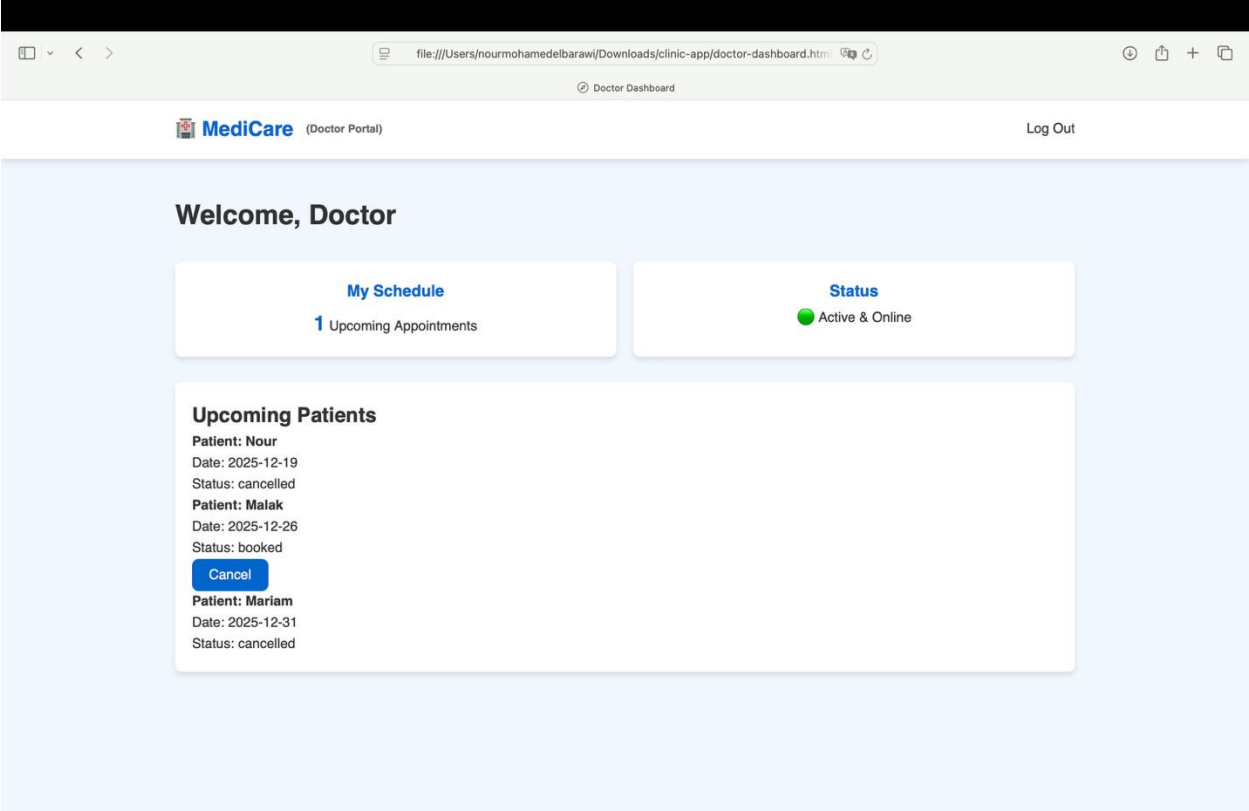


### Patient's Appointment Cancellation:

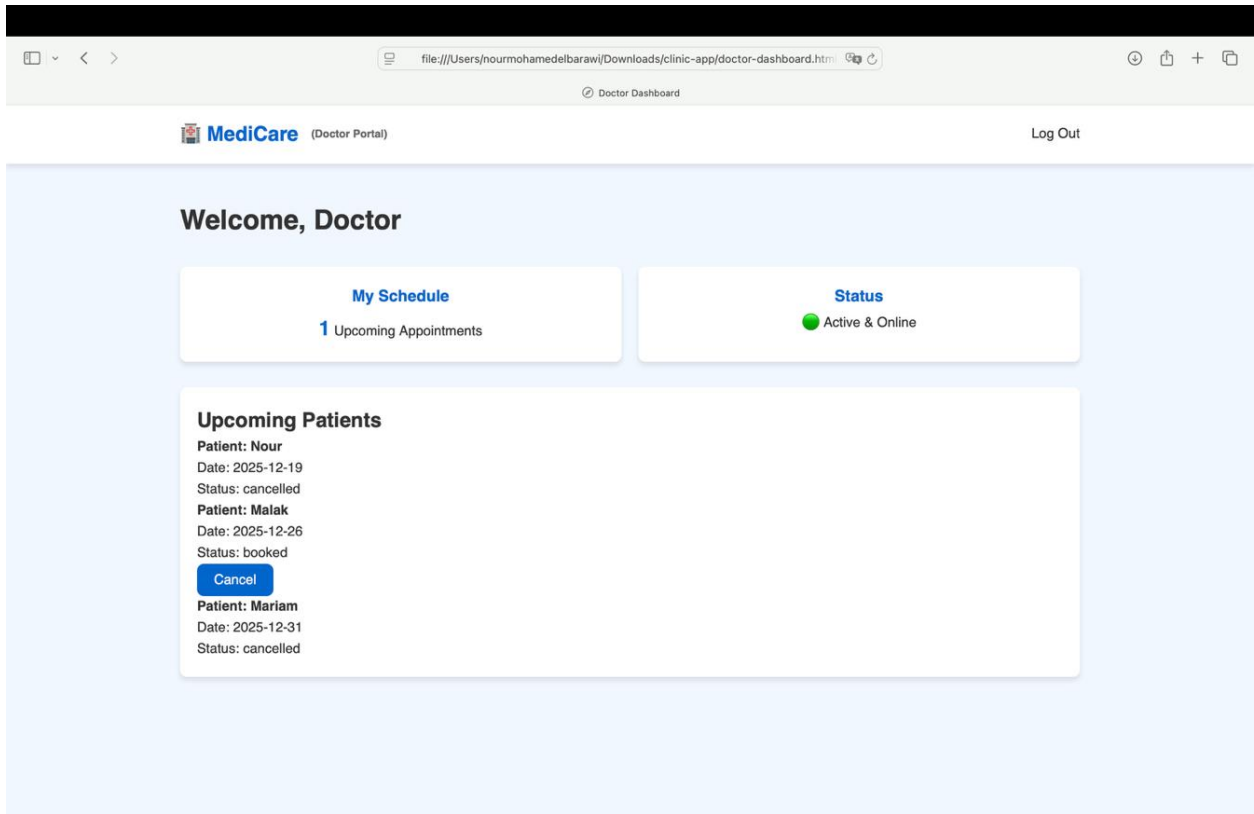
**Safety Mechanisms:** *Double-Confirmation Flow*; a significant design choice was the implementation of a Confirmation Modal for deletions. This forces a "stop-and-think" moment for the user, requiring a second click before an appointment is removed from the database. *whats*



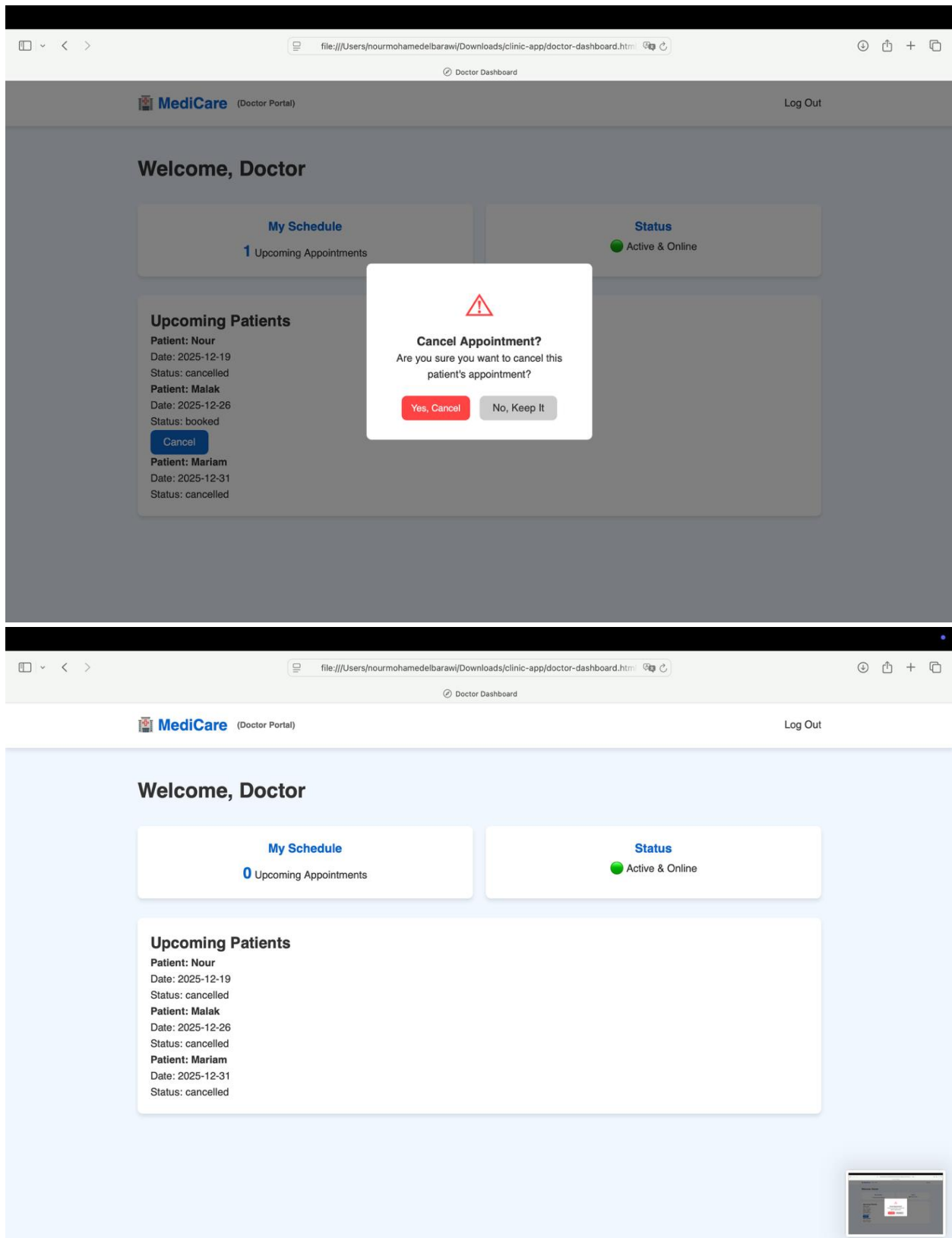
**Appointment Canceled in Doctor's Dashboard:**



Doctor’s Viewing Point:



***Doctor Cancelling Appointments:***



**Appointment Canacted from patient Dashboard:**

My Dashboard

MediCare

Log Out

Welcome, Patient

My Appointments

0 Upcoming

Book New

Medical History

No history available.

Profile Settings

Update your email or password.

Edit Profile

Your Schedule

Dr. Nour Mohamed

Pediatrician

Date: 2025-12-26

Status: cancelled

New DatabaseOpen DatabaseWrite ChangesRevert ChangesUndoOpen ProjectSave ProjectAttach DatabaseClose Database

Database StructureBrowse DataEdit PragmasExecute SQL

Table: appointments

| id  | patient_id | doctor_id | appointment_date | status    | created_at          |
|-----|------------|-----------|------------------|-----------|---------------------|
| ... | Filter     | Filter    | Filter           | Filter    | Filter              |
| 1   | 1          | 1         | 2025-12-20       | cancelled | 2025-12-15 17:26:50 |
| 2   | 2          | 3         | 2025-12-19       | cancelled | 2025-12-15 18:57:13 |
| 3   | 3          | 3         | 2025-12-26       | cancelled | 2025-12-15 19:01:19 |
| 4   | 4          | 2         | 2025-12-19       | cancelled | 2025-12-15 19:39:44 |
| 5   | 4          | 4         | 2025-12-22       | cancelled | 2025-12-15 19:42:15 |
| 6   | 6          | 3         | 2025-12-31       | cancelled | 2025-12-15 20:23:00 |